

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

This document evaluates the quality of the Rising Waters codebase (app.py, train_model.py, predict.py) in terms of structure, readability, and reusability.

Code Layout Checklist

S. No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing used throughout the code	Yes	PEP-8 style 4-space indentation across all files
2	Proper File Structure	Files and folders are logically organized	Yes	Separate data/, models/, static/, templates/ folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g., X_train, y_train, prediction, values
4	Function / Method Names	Functions are descriptively named	Yes	predict_flood(), home(), predict()
5	Code Comments	Inline and block comments explain logic	Partial	train_model.py is commented; app.py/predict.py are minimal
6	Modular Design	Code is split into reusable functions/modules	Yes	Prediction logic isolated in predict.py, separate from app.py
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate logic found across modules
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic type conversion only; explicit try/except can be added

Reusable Components / Modules

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	predict_flood() function	Python	Called by app.py for every	High

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			prediction request	
2	train_model.py pipeline	Python	Re-run independently to retrain the model on updated datasets	High
3	templates/index.html	HTML / Jinja2	Reused for both the input form and the result display	Medium
4	static/style.css	CSS	Shared styling across the single-page application	Medium

Overall Code Quality Assessment

Aspect	Rating (1–5)	Comments
Code Layout & Structure	4	Clear separation of concerns across app.py, train_model.py, and predict.py
Readability	4	Descriptive variable and function names throughout
Reusability	4	Prediction logic decoupled from Flask routing for easy reuse
Documentation / Comments	3	README is thorough; in-code comments could be expanded
Overall Score	4	Well-structured for a small ML web application